Anirudh Chandan

Backend Software Engineer · Node.js · Python · Distributed Systems

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SUMMARY

Backend software engineer (2+ years) building **scalable REST APIs, microservices, and real-time systems** in Node.js, Python, and Java (Spring Boot). Production experience with **Kafka event pipelines (exactly-once processing), Redis caching and distributed locks, PostgreSQL/MySQL, WebSockets, and AWS/GCP**. Built **PyDB, a database storage engine from scratch** (B-Tree indexes, write-ahead log, crash recovery). Strong CS fundamentals: 450+ LeetCode, system design.

EXPERIENCE

Independent Software Contractor *Aug 2026 – Present*
Backend / Full-Stack Engineer (freelance) *Remote · India / Australia*

- Built a **multi-tenant document workflow platform** (Python/FastAPI, PostgreSQL row-level security, Celery) for a shipping-management client: layered document classification, **YAML cross-document validation rules** (e.g. bill-of-lading weights vs manifest ±0.5%), maker-checker review, immutable versioning, 400+ automated tests; generalised into pluggable industry packs (shipping, procure-to-pay).
- Delivered a **live AI-CCTV alert dashboard** (Node.js) for a gated villa community: ingests events from an on-prem AI box via HTTP callbacks, patrol-compliance timers, EN/HI voice alerts, role-based remote access, Excel/PDF reports — **in production since Sep 2026**.

Docplix *Nov 2025 – Present*
Software Engineer *Remote*

- Built and led 30+ Node.js/Sequelize REST APIs for an EHR platform serving **2,300+ clinics**, as one of **4 engineers**; ran a **zero-downtime v1→v2 migration with 100% data integrity**.
- Designed **optimistic concurrency control and distributed Redis locks** for the central sync service, eliminating race conditions behind concurrent-write data mismatches.
- Replaced legacy polling with a push notification system (**WebSockets + Redis Pub/Sub**), cutting backend traffic by 80%.

Genpact *Feb 2024 – Oct 2025*
Software Engineer (promoted from Intern) *Bengaluru, India*

- Engineered **idempotent, exactly-once processing** for a Kafka tax-ingestion pipeline using database constraints, guaranteeing correctness across partitions during network failures.
- Built resilient Python (Flask) microservices with automated retries and dead-letter queues, serving 24 modules with **zero data loss**.
- Integrated structured logging and **distributed tracing** across serverless GCP Cloud Functions, cutting MTTR for critical production bugs by **40%**.

PROJECTS

PyDB — Database Storage Engine Built From Scratch. A disk-based transactional storage engine in pure Python: its own **pager and 8 KB page cache**, raw binary serialization, **clustered and secondary B-Tree indexes**, and a **write-ahead log** with redo recovery and atomic checkpoints. Recovery is tested by **SIGKILLing a writer mid-workload**; CI on every push. Also ported to TypeScript, running **live in the browser**. (live demo · code)

ProjAuto — Multi-tenant SaaS ERP (20+ live tenants, 120+ services). Top contributor (711 of 1,100 commits): migrated a legacy Java monolith to **Spring Boot + React**, eliminated N+1 queries across 53 endpoints (**15.6s → 2.4s**), built a distributed token-bucket rate limiter, and closed account-takeover and cross-tenant IDOR security holes.

SKILLS

Backend: Node.js (Express, Sequelize), Python (Flask, FastAPI), Java (Spring Boot), TypeScript, REST API design, microservices architecture, WebSockets / real-time systems

Data & Messaging: PostgreSQL (incl. row-level security), MySQL, Redis (caching, Pub/Sub, distributed locks), Apache Kafka, BullMQ, Celery, database internals (B-Trees, write-ahead logging, ACID)

Cloud & DevOps: AWS, GCP (Cloud Functions / serverless), Docker, Linux, Git, CI/CD (GitHub Actions)

Practices: system design, distributed systems, concurrency control, rate limiting, performance optimization, testing (Vitest, pytest), distributed tracing, structured logging

Foundations: Data Structures & Algorithms (450+ LeetCode), OOP, SQL

EDUCATION & ACHIEVEMENTS

Graphic Era University, Dehradun *2020 – 2024*

B.Tech, Computer Science & Engineering — CGPA 9.11

Technical blog on systems internals (anirudhchandan.com/blog).